

EDUCATION

Carnegie Mellon University

Visiting Scholar, School of Computer Science (S3D)

Pittsburgh, PA, USA

Sep. 2026–Feb. 2027

Seoul National University

Ph.D. in Intelligence and Information (Advisor: Prof. Nojun Kwak)

Seoul, Korea

Sep. 2024–Present

Ewha Womans University

B.S. in Computer Science and Engineering

Seoul, Korea

Mar. 2020–Aug. 2024

PUBLICATIONS

- [1] K. Kim, C. Kim, J. Shin, T. Kwon, J. Kim, **M. Koo**, and H. Park, “Phasor: Phase-anchored universal action representations for humanoid embodiments”, *arXiv preprint arXiv:2606.01851*, 2026. [link].
- [2] S. Panpatil, A. Singh, **M. Koo**, C. Kim, H. Park, and D. Choi, “Egosafetybench: A diagnostic egocentric video benchmark for evaluating embodied vlms as runtime safety guards”, *arXiv preprint arXiv:2607.00218*, 2026. [link].
- [3] J. Lee, **M. Koo**, and N. Kwak, “Csf: Black-box fingerprinting via compositional semantics for text-to-image models”, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 16 572–16 582. [link].
- [4] H. Lee*, **M. Koo***, Y. Song, and N. Kwak, “Targeted data protection for diffusion model by matching training trajectory”, in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 5854–5862. [link].
- [5] Y. Song, J. Lee, **M. Koo**, J. Lee, and N. Kwak, “Point-to-point: Sparse motion guidance for controllable video editing”, *arXiv preprint arXiv:2511.18277*, 2025. [link].
- [6] M. Park, C. Park, S. Lim, **M. Koo**, H. Lee, W. W. Ro, and S. Kim, “Adversarial purification via super-resolution and diffusion”, in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 4605–4615. [link].

* Equal contribution in [4].

RESEARCH EXPERIENCE

AIM Intelligence

AI Researcher

Seoul, Korea

Jan. 2026–Jul. 2026

– Project: Runtime Safeguard for Physical AI Agents

- Developed a vision-language model (VLM)-based runtime safeguard for physical AI agents, focusing on real-time detection and intervention of unsafe behaviors.
- Deployed the safeguard model on a Unitree G1 humanoid robot and validated real-time safety intervention in collaboration with an on-device engineer.
- Contributed to embodied AI safety research, including EgoSafetyBench and PHASOR.

Korea Institute of Science and Technology (KIST)

Undergraduate Researcher, AI & Robotics Institute

Seoul, Korea

Jun. 2023–Dec. 2023

- **Project: Adversarial Purification with Diffusion and Super-Resolution Models**
 - Developed adversarial defense methods using diffusion and super-resolution models for image purification.
 - Conducted comparative analysis of purification performance across multiple adversarial attack scenarios.

HONORS & AWARDS

IITP Government Fellowship — CMU AI Advanced Program

2026

Ministry of Science and ICT, Korea

SELECTED PROJECTS

A Single Word Bypass – How Name Tokens Break Data Protection [Paper]

Course Project, Seoul National University

Mar. 2025–Jun. 2025

- Identified a token-dependent failure mode in data protection methods for diffusion model personalization, showing that common name tokens can bypass existing safeguards.
- Conducted mechanistic analyses of parameter updates, cross-attention patterns, and text embeddings to characterize the source of protection failure.

CareSpoon: AI-based Nutrition Monitoring Service for Seniors

Capstone Design, Ewha Womans University

Feb. 2023–Jun. 2023

- Developed a PyTorch-based object detection pipeline for food recognition and deployed the inference system using FastAPI.
- Constructed and curated a Korean food image dataset from AI HUB for model training and evaluation.

Smart Bakery: Self-Order Kiosk Web Service

Personal Project

Jul. 2021–Aug. 2021

- Built an automated bakery checkout prototype combining a Vanilla JavaScript frontend with a PyTorch image classification backend.

TEACHING

Samsung Electronics

Teaching Assistant, AI Expert Program — Adversarial Attack & Defense

2024–2026

LEADERSHIP EXPERIENCE

SK LOOKIE

President, Social Venture Club

Feb. 2023–Feb. 2024

AI Club Euron

Manager / Academic Lead — Computer Vision study and technical seminars

Mar. 2021–Aug. 2022